

Anirudh Som

Contact Info:

Email: contact.anirudh.som@gmail.com **Website** [Website](#) **LinkedIn** [LinkedIn](#) **Google Scholar** [Google Scholar](#)

Work Authorization: U.S. Permanent Resident

Skills:

Programming – Python, Matlab, XML, Latex

Libraries & Frameworks – PyTorch, TensorFlow, Transformers, Scikit-learn, Pandas, OpenCV, DeepSpeed

LLMs & VLMs – GPT, LLaMa, Mixtral, Mistral, Open LLaMa | RAG, RLHF, SFT, LoRA, PEFT, In-Context Learning

Evaluation Benchmarks – HELM, MMLU, TruthfulQA, BLEU, ROUGE, BERTScore, CIDEr

Infrastructure – Docker, Kubernetes, Gitlab, Artifactory, AWS, GPU cluster training, FSDP, ZeRO

Data Handling – Document parsing, OCR pipelines, Structured extraction from unstructured sources

Education:

Ph.D., Electrical Engineering – Arizona State University, Tempe, AZ

2020

Professional Experience:

• **Vice President - Applied AI/ML** – JPMorganChase, New York, NY

2025 – Present

• **Advanced Computer Scientist** – SRI International, Princeton, NJ

2021 – 2025

- **CV & NLP Systems:** Architected and led deployment of transformer-based CV and LLM pipelines for real time and offline video understanding; enabled scene segmentation, summarization, metadata extraction, and culturally aware emotion recognition in HMIs.
- **Document Intelligence:** Developed OCR+NLP pipelines for scanned inputs and LLM systems for extracting structured insights from unstructured medical and educational documents.
- **Mentorship & Leadership:** Acted as technical lead for multiple federal and commercial AI projects. Supervised 2-4 junior ML engineers and interns across project lifecycles.
- **Commercial Use Cases:** Delivered AI-based prototypes and proof-of-concepts to Fortune 500 companies like L'Oréal and Honda.
- **Product Collaboration:** Partnered with cross-department teams to deploy and maintain ML solutions, aligning technical efforts with product goals and ensuring operational reliability.

Key Projects:

- **ARPA-H PARADIGM:** CV+LLM-driven task guidance system for rural healthcare support.
- **DARPA CCU:** CV+LLM pipelines for cross-cultural awareness in dialogue systems.
- **DARPA EDGE:** Real-time emotion-aware HMI systems.
- **NSF:** Automated classroom collaboration assessment using multimodal architectures.

Pre-Doctoral Experience:

• **Research Intern** – SRI International, Princeton, NJ

2020

Multimodal deep learning for educational assessment.

• **Digital Pathology Intern** – Roche Diagnostics, Santa Clara, CA

2019

Image segmentation, federated learning in digital pathology.

• **Computation Intern** – Lawrence Livermore National Laboratory, Livermore, CA

2018

Time-series deep learning for heart signal modeling.

• **Graduate Intern** – Mayo Clinic, Phoenix, AZ

2015

ML for cancer detection in CT/MRI.

Selected Publications:

- [1] **A. Som** et al. "RAG-based Social Norm Detection in Multi-cultural Conversation Settings". Under Review, 2025.
- [2] **A. Som** et al. "[Demonstrations Are All You Need: Advancing Offensive Content Paraphrasing using In-Context Learning](#)". Accepted at the Association for Computational Linguistics (ACL) Findings, 2024.
- [3] **A. Som**, et al. "[Automated Student Group Collaboration Assessment and Recommendation System Using Individual Role and Behavioral Cues](#)". Accepted at the Frontiers in Computer Science Journal, 2021.
- [4] **A. Som**, et al. "[Perturbation Robust Representations of Topological Persistence Diagrams](#)". Accepted at the European Conference on Computer Vision (ECCV), 2018.

Complete List: [Google Scholar](#)

Patents:

- [1] [Machine Learning Model Prompt Demonstration Selection.](#)
- [2] [Automated Identification of Necrotic Regions in Digital Images of Multiplex Immunofluorescence-Stained Tissue.](#)
- [3] [Federated Learning System for Training Machine Learning Algorithms Without Accessing Patient Data to Protect Patient Privacy.](#)
- [4] [Automated Collaboration Skills Assessment.](#)

Awards:

- "Research Recognition Award" – CVT Group at SRI International
- "Facilitators' Choice Award" - 2021 NSF STEM for All Video Showcase

2022 – 2024

2021